EU4HEALTH PROGRAMME

EU4Health Project Grant

Equitable AI-Assisted Colorectal Cancer Screening Across Regions

CANCERSCREEN-EU

Field	Value
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Abstract

Colorectal cancer (CRC) remains the second leading cause of cancer death in the European Union, yet screening participation rates fall below 25% in many disadvantaged regions — far short of the targets set by the 2022 Council Recommendation on cancer screening and Europe's Beating Cancer Plan. CANCERSCREEN-EU pilots, evaluates, and provides scale-up guidance for an equitable, AI-assisted CRC screening pathway in underserved regions across Italy, Romania, Portugal, and Lithuania over 42 months.

The project co-designs an integrated FIT-based screening pathway incorporating a prospectively validated AI decision-support module for triage and colonoscopy prioritisation. The pathway is deployed across eight regional pilot sites, actively enrolling a minimum of 40,000 eligible individuals aged 50–74 through multilingual, culturally adapted community outreach strategies and mobile screening units. A rigorous mixed-methods evaluation — disaggregated by gender, socio-economic status, and geographic remoteness — assesses screening uptake, adenoma detection rates, AI performance, and equity outcomes against pre-specified targets, including a primary target of at least a 15-percentage-point increase in screening participation relative to regional baselines.

The consortium — led by the National Cancer Institute of Italy and including national health authorities and oncology institutes in Romania, Portugal, and Lithuania — produces a transferable, evidence-based Scale-Up Playbook validated with regional health authorities, and secures at least six formal replication commitments enabling post-project roll-out without additional EU grant funding. All outputs are aligned with EHDS interoperability standards, EU AI Act requirements, and FAIR Open Science principles.

CANCERSCREEN-EU directly addresses cancer inequalities within the EU, advances the responsible integration of AI in public health screening, and generates the real-world evidence base needed to support the wider national roll-out of equitable CRC screening programmes across Europe.
Keywords: colorectal cancer screening, AI decision support, health equity, Europe's Beating Cancer Plan, cancer prevention, FIT-based screening, underserved regions, implementation science, EU4Health, digital health

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SECTION 1 - RELEVANCE

1.1 Objectives and Ambition

Colorectal cancer (CRC) is the second leading cause of cancer death in the European Union, with five-year survival rates varying more than twofold across Member States. Participation in organised CRC screening programmes remains markedly unequal: underserved rural and socio-economically deprived regions in countries such as Romania, Lithuania, and Portugal record participation rates below 25%, far beneath the targets established in the 2022 Council Recommendation on cancer screening. CANCERSCREEN-EU addresses this critical gap by piloting, evaluating, and providing scale-up guidance for an equitable, AI-assisted CRC screening pathway tailored to the structural realities of underserved regions across four EU Member States.

The primary objective of CANCERSCREEN-EU is to increase CRC screening participation rates by at least 15 percentage points in targeted underserved regions by month 42, relative to regional baselines established at project inception. Secondary objectives are: (i) to validate and deploy a CE-marked AI decision-support tool for faecal immunochemical test (FIT) result triage and colonoscopy prioritisation; (ii) to design and test culturally and linguistically adapted community outreach strategies that measurably reduce equity gaps in screening uptake; (iii) to produce a transferable, evidence-based scale-up playbook for wider EU adoption, validated with at least six regional health authorities; and (iv) to strengthen national cancer registry linkage and real-world evidence generation capacity in participating countries. All objectives are Specific, Measurable, Achievable, Relevant, and Time-bound (SMART), tracked through the project's dedicated monitoring framework, and directly aligned with the EU Cancer Screening Scheme and the Europe's Beating Cancer Plan milestone of ensuring that 90% of eligible EU citizens are offered breast, cervical, and colorectal cancer screening by 2025–2030.

CANCERSCREEN-EU goes beyond the current state of the art in three important respects. First, whereas existing EU-funded screening pilots have predominantly focused on high-capacity health systems, this project deliberately grounds its design and implementation in lower-capacity Member States (Romania and Lithuania) where health system constraints are most acute and equity gaps are widest. Second, the AI decision-support module is not merely adapted from existing tools but prospectively validated against European colonoscopy datasets in multiple national regulatory contexts, generating novel real-world performance evidence that is currently absent from the literature. Third, the project's mixed-methods evaluation framework explicitly quantifies equity outcomes — disaggregated by gender, socio-economic status, and geographic remoteness — providing an evidence base that is systematically missing from prior CRC screening research in Europe.

1.2 Relation to the Work Programme

CANCERSCREEN-EU is directly responsive to the EU4Health Programme call for proposals under EU4H-2026-SANTE-PJ, which explicitly invites actions to pilot and implement cancer screening programmes aligned with the 2022 Council Recommendation on strengthening prevention through early detection (OJ C 473, 13.12.2022). The project implements the EU4Health Programme's general objective of improving and fostering health in the Union to reduce the burden of non-communicable diseases (Article 3(a), Regulation (EU) 2021/522) through the specific objectives defined in Article 4, points (a), (g), and (i) of that Regulation — namely: promoting health and preventing disease; supporting actions to address health inequalities; and fostering digital transformation in health and care.

The action builds directly on the Joint Action EUCanScreen and the Cancer Image Europe platform under the Digital Europe Programme, ensuring complementarity and avoiding duplication of effort. It

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reinforces the planned Commission monitoring activities on the third EU Cancer Screening Monitoring Report by contributing harmonised, equity-disaggregated data from four Member States. The consortium's focus on Romania and Lithuania — countries whose GNI per inhabitant is below 90% of the EU average — ensures that more than 30% of the project budget is directed to lower-income Member States, qualifying the project for the 80% exceptional co-financing rate under EU4Health rules.

CANCERSCREEN-EU further aligns with the European Health Data Space (EHDS) Regulation by adopting interoperable data standards and FAIR (Findable, Accessible, Interoperable, Reusable) principles for all project-generated datasets. The project thus constitutes a coherent, additive contribution to the EU policy architecture on cancer, digital health, and health equity — rather than a standalone initiative — and its outputs are explicitly designed to feed into EU-level guidance through the Joint Action EUCanScreen.

1.3 Concept and Methodology

The CANCERSCREEN-EU methodology is grounded in implementation science and follows a stepwise, evidence-based approach to screening programme scale-up. The project deploys a mixed-methods design that combines quantitative evaluation of screening outcomes with qualitative assessment of equity barriers and community engagement processes. The conceptual model is structured around three interconnected phases: (1) pathway co-design and AI tool contextualisation; (2) regional pilot implementation; and (3) cross-site evaluation and scale-up guidance.

In Phase 1 (months 1–12), the consortium will map existing CRC screening infrastructure, equity gaps, and AI readiness across all four partner countries, co-designing an adapted FIT-based screening pathway with AI-assisted triage. The AI decision-support module — built on a validated convolutional neural network architecture trained on European colonoscopy datasets — will be contextualised to national regulatory and clinical environments. Prospective clinical validation will be conducted at two sites (Italy and Portugal) prior to pilot deployment, generating the real-world performance evidence required for regulatory deployment in participating countries.

In Phase 2 (months 10–36), regional pilots will be launched in at least eight underserved regions across Romania, Lithuania, Portugal, and Italy, each enrolling a minimum of 5,000 eligible individuals aged 50–74. Community health mediators, multilingual invitation materials, and mobile screening units will be deployed in each region to systematically address structural access barriers identified in Phase 1. The overlapping start of Phases 1 and 2 (from month 10) reflects a pragmatic, adaptive design: sites completing landscape assessment and training early will begin recruitment without waiting for lagging sites, maximising the recruitment window within the project's 42-month duration.

In Phase 3 (months 30–42), a rigorous mixed-methods evaluation will assess screening uptake, adenoma detection rates, equity outcomes, and AI performance metrics against pre-specified thresholds. Quantitative analyses will use interrupted time-series and difference-in-differences methods where pre-intervention data are available. Qualitative analyses will employ framework analysis to synthesise findings from focus groups and key informant interviews with community health mediators, clinicians, and patients. All methods comply fully with GDPR, the EU AI Act's transparency and human oversight requirements for high-risk AI systems, and applicable clinical ethics standards. The gender dimension is embedded throughout: all analyses are sex-disaggregated and community outreach strategies are explicitly tailored to address documented gender differences in CRC screening participation.

1.4 Interdisciplinary Considerations

CANCERSCREEN-EU integrates cross-cutting dimensions of equity, digital transformation, and ethical governance as structural elements of its design rather than peripheral considerations.

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On health equity, the project adopts a health-equity-in-all-policies lens. All pilot sites include targeted sub-analyses disaggregated by gender, age cohort, socio-economic status, and geographic remoteness. Community outreach strategies are co-designed with patient advocacy groups and local civil society organisations from month 6 onwards, ensuring cultural acceptability, linguistic accessibility, and community trust — factors that systematic reviews consistently identify as primary determinants of screening uptake in underserved populations.

On digital transformation, the AI decision-support tool is developed and deployed in strict accordance with the EU AI Act's requirements for high-risk medical AI systems, including full auditability, explainability, and mandatory human oversight provisions. The project contributes to EHDS interoperability by adopting HL7 FHIR standards for data exchange between screening registries and clinical systems across all four partner countries, generating a replicable technical blueprint for future cross-border health data integration.

On gender and social inclusion, the consortium applies the European Institute for Gender Equality (EIGE) gender mainstreaming methodology to all communication materials, community engagement strategies, and workforce training programmes, ensuring that well-documented gender differences in CRC screening participation are explicitly and systematically addressed rather than assumed away.

On interdisciplinary collaboration, the consortium deliberately integrates expertise in oncology, public health, implementation science, medical AI, health data governance, health economics, and community engagement. This disciplinary breadth is essential because no single domain can address the multi-layered barriers — clinical, technological, social, and regulatory — that perpetuate screening inequity in underserved regions.

Ethical oversight is provided by the coordinating institute's independent ethics board, with a consortium-level Data Protection Officer appointed from month 1. The project's ethics framework covers informed consent procedures, data minimisation, algorithmic fairness assessment, and the governance of AI-generated clinical recommendations.

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SECTION 2 - IMPACT

2.1 Pathways Towards Impact

The direct beneficiaries of CANCERSCREEN-EU are approximately 40,000 eligible individuals aged 50–74 residing in targeted underserved regions across Italy, Romania, Portugal, and Lithuania, who will be actively invited to participate in the AI-assisted CRC screening pathway during the pilot phase. Indirect beneficiaries include national health systems — which will gain a validated, ready-to-deploy scale-up playbook — and the broader EU population, through the medium- and long-term reductions in cancer-related morbidity and mortality that wider adoption of equitable screening pathways is projected to deliver.

In the short term (within the project lifetime), CANCERSCREEN-EU will generate measurable increases in CRC screening participation rates, improved adenoma detection through AI-assisted triage, and strengthened regional health workforce capacity in screening delivery, documented through the project monitoring framework and interim evaluation report (D4.1, month 30).

In the medium term (two to five years post-project), national health authorities in all four countries will have access to a validated scale-up playbook and replication toolkit (D4.3), enabling systematic roll-out to additional regions without further EU grant funding. The consortium's strategy of securing at least six formal replication commitments from regional health authorities by month 42 creates a concrete institutional foundation for this medium-term uptake pathway, moving beyond the dissemination of findings towards binding commitments to act on them.

In the long term (five to ten years), widespread adoption of the CANCERSCREEN-EU pathway model is projected to contribute to a measurable reduction in late-stage CRC diagnoses and associated mortality in participating Member States, directly supporting the Europe's Beating Cancer Plan target of saving 3 million lives by 2030. The pathway to long-term impact is further operationalised through active engagement with the Joint Action EUCanScreen to embed CANCERSCREEN-EU findings in EU-level guidance documents, ensuring that the project's evidence base influences policy and practice beyond the four partner countries and beyond the project's own timeline.

The project's impact pathway is realistic and evidence-based: a 15-percentage-point increase in screening participation in eight underserved regions enrolling 40,000 individuals, combined with AI-assisted improvements in adenoma detection, is consistent with the best available evidence from comparable European screening pilots and with the capacity of the consortium partners to deliver.

2.2 Dissemination, Exploitation and Communication

CANCERSCREEN-EU will implement a structured, audience-differentiated communication, dissemination, and exploitation strategy governed by Work Package 5 (led by the Portuguese Oncology Institute) and active throughout the full 42-month project lifetime.

The primary target audiences are: national health authorities and cancer screening programme managers (primary policy audience); clinical professionals including gastroenterologists, general practitioners, and nurses (primary practice audience); patient organisations and community representatives (primary public engagement audience); EU-level policymakers and DG SANTE (primary EU policy audience); and the scientific community (primary academic audience). Each audience receives tailored content and communication channels.

Dissemination activities include: a publicly accessible, multilingual project website launched by month 3; a minimum of eight peer-reviewed publications in open-access journals, covering AI validation results, equity analyses, implementation science findings, and the scale-up evidence base;

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presentations at the European Cancer Congress and the European Society of Gastrointestinal Endoscopy (ESGE) Annual Meeting; four policy briefs addressed to national ministries of health and DG SANTE (D5.2, month 38); and a final public conference in Brussels (month 40) convening representatives of all four national health authorities, European Commission officials, and patient organisations.

Exploitation activities are designed from the outset to ensure sustainability beyond the grant period. The AI decision-support tool will be made available as an open-source module under a permissive open-source licence, enabling adaptation and deployment by health authorities in other Member States without licensing costs. The scale-up playbook will be formally submitted to the Joint Action EUCanScreen for integration into EU-level guidance, creating an institutionalised channel for the project's evidence to shape practice across the EU. All training materials and clinical protocols will be deposited in the EU Health Knowledge Gateway, ensuring long-term public accessibility. Patient advocacy organisations will be engaged as co-designers of communication materials from month 6 and as community ambassadors throughout the pilot phase, building the community trust that is a prerequisite for sustained screening participation beyond the project's funding period. Open Science principles are applied throughout: all datasets will be published in FAIR-compliant repositories and all publications will be made open-access in compliance with Plan S requirements.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
CRC screening participation rate in pilot regions	Increase by ≥15 percentage points from regional baseline by month 42	Administrative screening registry data; monthly monitoring reports per pilot site
Adenoma detection rate (ADR) in AI-assisted vs. standard pathway	≥10% relative improvement in ADR in AI-assisted arm	Endoscopy unit records; blinded evaluation by clinical evaluation committee
Equity gap in screening uptake (lowest vs. highest socio-economic quartile)	Reduction of equity gap by ≥20% relative to baseline	Screening uptake data disaggregated by socio-economic indicators; annual equity analysis report
Number of community outreach actions implemented	Minimum 48 community outreach events across 8 pilot regions by month 36	Activity logs and attendance records compiled by regional coordinators
Number of regional/national health authorities committing to replication	≥6 formal replication commitments by month 42	Signed letters of intent from health authorities; recorded in final evaluation report
Open-access scientific publications	≥8 peer-reviewed publications by month 42	Publication database maintained by WP5 lead

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SECTION 3 - QUALITY OF IMPLEMENTATION

3.1 Work Plan and Work Packages

WP	Title	Lead	PM	Duration
WP1	Project Management and Coordination	National Cancer Institute Italy (INCa-IT)	18	M1-M42
WP2	Pathway Co-Design and AI Tool Contextualisation	National Cancer Institute Italy (INCa-IT)	52	M1-M18
WP3	Regional Pilot Implementation	Romanian National Health Authority (RNHA-RO)	88	M10-M36
WP4	Monitoring, Evaluation, and Scale-Up Guidance	Lithuanian University of Health Sciences (LUHS-LT)	44	M10-M42
WP5	Dissemination, Communication, and Stakeholder Engagement	Portuguese Oncology Institute (IPO-PT)	30	M1-M42

WP1: Project Management and Coordination

Lead: National Cancer Institute Italy (INCa-IT) | Person-months: 18 | Duration: M1 - M42

WP1 ensures the overall scientific, administrative, and financial coordination of the CANCERSCREEN-EU consortium across its full 42-month duration. It establishes and maintains governance structures — including the Project Management Board, Scientific Advisory Board, and Ethics Oversight Committee — manages reporting obligations to HaDEA, oversees consortium agreement compliance, and implements the project's risk and quality management frameworks. WP1 serves as the central coordination hub, facilitating information flow between work packages and ensuring timely resolution of implementation challenges. Its outputs are preconditions for the effective delivery of all other WPs.

T1.1: Governance and legal setup - Establish and operationalise the consortium agreement, Project Management Board, Scientific Advisory Board, and Ethics Oversight Committee by month 2. Define decision-making procedures, intellectual property arrangements, and data governance responsibilities. Ensure that ethics approval from the coordinating institution's independent ethics board is obtained and submitted to HaDEA prior to any data collection activity.

T1.2: Financial and administrative management - Coordinate all financial management activities across the consortium, including budget monitoring, internal financial reporting, audit preparation, and timely submission of periodic progress and final reports to HaDEA in accordance with grant agreement requirements. Maintain a consolidated financial dashboard accessible to all partners.

T1.3: Quality assurance and risk monitoring - Implement and maintain the project risk register, updated quarterly in line with the risk management framework. Conduct internal quality reviews at months 14, 28, and 40, covering scientific output quality, implementation progress against milestones, and compliance with Open Science and data management obligations. Escalate identified risks to the Project Management Board with recommended mitigation actions.

WP2: Pathway Co-Design and AI Tool Contextualisation

Lead: National Cancer Institute Italy (INCa-IT) | Person-months: 52 | Duration: M1 - M18

WP2 lays the scientific and operational foundation for the pilot phase by generating the contextualised evidence base and validated tools required for equitable deployment. It maps existing CRC screening infrastructure, equity gaps, and AI regulatory readiness across all four partner countries; co-designs — with clinical partners and patient representatives — the integrated FIT-based AI-assisted screening pathway; and conducts prospective clinical validation of the AI decision-support module at two sites (Italy and Portugal) prior to any pilot deployment. WP2 outputs are critical dependencies for WP3: no pilot site will begin active recruitment until the pathway protocol and AI validation report have been reviewed and approved by the Scientific Advisory Board.

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T2.1: Country-level screening landscape mapping - Conduct structured, standardised assessments of CRC screening programme status, existing invitation and follow-up infrastructure, AI readiness, and equity barriers in Italy, Romania, Portugal, and Lithuania. Assessments will use a harmonised template drawing on the EUCanScreen monitoring framework, supplemented by key informant interviews with national screening programme managers. Four national landscape assessment reports will be produced and approved by the Scientific Advisory Board by month 9 (D2.1).
T2.2: AI decision-support tool contextualisation and validation - Contextualise the AI FIT triage module to the national regulatory (EU AI Act, MDR) and clinical environments of Italy and Portugal. Conduct prospective clinical validation at two sites against pre-specified performance thresholds (sensitivity, specificity, and positive predictive value for advanced adenoma detection). Produce a validation report (D2.2) with full statistical performance metrics by month 15. Engage national competent authorities in Romania and Lithuania from month 3 to anticipate and resolve regulatory requirements in advance of pilot deployment.
T2.3: Integrated pathway design and training - Co-design the integrated AI-assisted screening pathway protocol with clinical partners, patient advocacy representatives, and community health mediator networks, ensuring that equity and gender mainstreaming considerations are embedded in invitation, triage, and follow-up procedures. Develop and deliver standardised training for regional healthcare teams across all eight pilot sites. Produce the Integrated Screening Pathway Protocol and Training Materials (D2.3) by month 17.

WP3: Regional Pilot Implementation

Lead: Romanian National Health Authority (RNHA-RO) | Person-months: 88 | Duration: M10 - M36

WP3 implements the co-designed, AI-assisted CRC screening pathway across eight underserved pilot regions in Italy, Romania, Portugal, and Lithuania, constituting the core operational phase of CANCERSCREEN-EU. Drawing on the pathway protocol and training materials produced in WP2, WP3 deploys community outreach strategies — including community health mediators, multilingual invitation campaigns, and mobile screening units — and enrolls a minimum of 40,000 eligible individuals aged 50–74 across the 27-month recruitment window (months 10–36). The Romanian National Health Authority leads this WP in recognition of Romania's position as the partner country with the most acute equity gaps and the greatest potential for measurable impact. All implementation activities generate real-time data fed into the WP4 monitoring framework.

T3.1: Pilot site preparation and recruitment infrastructure - Establish fully operational invitation systems, mobile screening unit deployment protocols, FIT kit distribution logistics, and community health mediator networks at all eight pilot sites by month 18 (MS4). Site readiness will be assessed against a standardised checklist developed in WP2 before any active recruitment commences. Contingency protocols for sites experiencing setup delays will be activated by the WP3 lead in coordination with WP1.
T3.2: Active screening invitation and community outreach - Implement multilingual, culturally adapted screening invitation campaigns and a minimum of 48 community outreach events — six per pilot region — across all eight pilot regions by month 36. Invitation materials will be co-designed with patient organisations and community health mediators (building on WP2 co-design activities) to maximise cultural acceptability and trust. Attendance and participation data from all community outreach events will be recorded in the WP4 monitoring system.
T3.3: AI-assisted FIT triage and colonoscopy referral - Deploy the validated AI decision-support module within routine screening workflows at all pilot sites following confirmation of site readiness (T3.1) and AI regulatory clearance in each national context. Monitor AI performance metrics — including sensitivity, specificity, and referral rates — on a monthly basis throughout the recruitment period, reporting to the WP4 monitoring framework. In sites where regulatory clearance is delayed, operate the FIT-based pathway using standard triage protocols as pre-specified in the fallback design, preserving primary screening uptake objectives.

WP4: Monitoring, Evaluation, and Scale-Up Guidance

Lead: Lithuanian University of Health Sciences (LUHS-LT) | Person-months: 44 | Duration: M10 - M42

WP4 provides the evidential backbone of CANCERSCREEN-EU, conducting rigorous mixed-methods evaluation of pilot outcomes and translating evaluation findings into actionable, transferable guidance for wider EU adoption. It implements the project monitoring framework from month 10, collecting harmonised outcome data across all eight pilot sites and producing six-monthly monitoring reports to inform adaptive management. Its evaluation methodology combines interrupted time-series analysis of screening KPIs with qualitative framework analysis of equity and community engagement outcomes. WP4's primary legacy outputs — the Interim Evaluation and Equity Analysis Report (D4.1), the Final Mixed-Methods Evaluation Report (D4.2), and the Scale-Up Playbook and Replication Toolkit (D4.3) — constitute the principal knowledge products through which CANCERSCREEN-EU will influence policy and practice beyond the consortium and beyond the grant

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period.
T4.1: Monitoring framework and data collection - Design and implement the project monitoring framework by month 10, including a harmonised data collection template, data quality assurance protocols, and a secure data management infrastructure compliant with GDPR and EHDS interoperability standards. Collect screening outcome data from all pilot sites on a monthly basis and produce six-monthly monitoring reports for the Project Management Board, identifying deviations from targets and recommending corrective actions.
T4.2: Mixed-methods impact evaluation - Conduct the mixed-methods evaluation of pilot outcomes across three analytical streams: (i) quantitative analysis of screening participation rates, adenoma detection rates, and equity outcomes using interrupted time-series and difference-in-differences methods; (ii) AI performance assessment against pre-specified validation thresholds; and (iii) qualitative assessment of equity barriers, community engagement effectiveness, and implementation fidelity through focus groups and key informant interviews. All analyses will be sex-disaggregated. Produce the Interim Evaluation Report (D4.1, month 30) and Final Evaluation Report (D4.2, month 41).
T4.3: Scale-up playbook and replication toolkit development - Synthesise quantitative and qualitative evaluation evidence into a transferable, practical Scale-Up Playbook and Replication Toolkit (D4.3, month 40) designed for use by national and regional health authorities across the EU. The playbook will cover pathway design, AI integration, community outreach, monitoring, and equity assurance. It will be validated through structured consultation with at least six regional health authorities (contributing to MS7) and submitted to the Joint Action EUCanScreen for integration into EU guidance documents.

WP5: Dissemination, Communication, and Stakeholder Engagement

Lead: Portuguese Oncology Institute (IPO-PT) | Person-months: 30 | Duration: M1 - M42

WP5 manages all external communication, scientific dissemination, stakeholder engagement, and open-access knowledge transfer activities throughout the project's 42-month lifetime, ensuring that CANCERSCREEN-EU's methods, findings, and tools are accessible to and actively used by EU policymakers, national health authorities, clinical communities, patient organisations, and the scientific community. WP5 operates from month 1 to build the visibility and stakeholder relationships needed to maximise uptake of project outputs and to secure the replication commitments that constitute a primary impact indicator. The Portuguese Oncology Institute leads WP5 given its established relationships with DG SANTE, the European Cancer Organisation, and patient advocacy networks across Southern and Eastern Europe.

T5.1: Communication strategy and project website - Develop and publish the project communication and dissemination strategy (D5.1) by month 3, covering audience segmentation, channel selection, messaging framework, and branding guidelines aligned with HaDEA visibility requirements. Launch a publicly accessible, multilingual project website by month 3, serving as the primary public-facing repository for project outputs, news, and resources throughout the project lifetime and for at least three years post-completion.
T5.2: Scientific dissemination and policy engagement - Coordinate the production and open-access publication of a minimum of eight peer-reviewed scientific articles covering AI validation, equity analyses, implementation science findings, and scale-up evidence. Represent the project at the European Cancer Congress and ESGE Annual Meeting. Produce four targeted policy briefs (D5.2) addressed to national ministries of health and DG SANTE by month 38, translating key project findings into actionable policy recommendations. Engage with the Joint Action EUCanScreen on an ongoing basis to position CANCERSCREEN-EU findings within EU guidance processes.
T5.3: Final public conference and sustainability engagement - Organise the final CANCERSCREEN-EU public conference in Brussels (month 40), convening representatives of all four national health authorities, European Commission officials, patient organisations, and the Joint Action EUCanScreen. The conference will present the scale-up playbook, showcase pilot results, and serve as the formal occasion for securing signed replication commitments from regional health authorities. Produce Final Conference Proceedings and a Replication Commitment Report (D5.3) by month 42, documenting the minimum six formal letters of intent secured from health authorities.

3.2 Consortium Composition

Organisation	Country	Type	Role	Description
National Cancer Institute Italy (INCa-IT)	Italy	National public research and clinical institute	Coordinator	Overall coordinator; leads WP1 and WP2; responsible for AI tool validation and scientific direction
Romanian National Health Authority	Romania	National public health authority	Partner	Lead WP3; coordinates pilot

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(RNHA-RO)				implementation in Romanian underserved regions; national registry linkage
Portuguese Oncology Institute (IPO-PT)	Portugal	National oncology institute	Partner	Lead WP5; co-leads AI validation (T2.2); coordinates Portuguese pilot sites and dissemination
Lithuanian University of Health Sciences (LUHS-LT)	Lithuania	Public academic health institution	Partner	Lead WP4; evaluation methodology; data governance; Lithuanian pilot coordination

The CANCERSCREEN-EU consortium has been assembled to combine scientific leadership with operational delivery capacity across the full spectrum of health system contexts in which CRC screening inequity is most acute. The coordinating institution — the National Cancer Institute of Italy — provides the scientific and AI validation expertise required to ensure methodological rigour and regulatory credibility. This scientific leadership is complemented by the implementation authority of national health authorities and oncology institutes in three additional Member States, two of which (Romania and Lithuania) have GNI per capita below 90% of the EU average and represent health systems where the equity gap in CRC screening participation is both widest and most consequential.

This combination is not accidental but strategically essential: a project led entirely by high-capacity institutions from Western Europe would lack the operational embeddedness needed to implement sustainable change in the regions where it is most needed. Conversely, a project without strong scientific coordination would lack the methodological rigour needed to generate evidence that is credible at EU level. CANCERSCREEN-EU achieves both.

The geographic spread across Southern Europe (Italy, Portugal), Eastern Europe (Romania), and Northern Europe (Lithuania) maximises the diversity of health system contexts covered and, consequently, the transferability of findings and tools to the widest possible range of EU Member States. All four partners have prior experience in EU-funded health projects, established cancer registry infrastructure, and active engagement with national cancer screening programmes — ensuring that the project builds on existing institutional capacity rather than constructing it from scratch. The consortium is of an appropriate size and composition to deliver the proposed work: large enough to cover the necessary geographic and disciplinary breadth, and focused enough to be managed efficiently within the proposed governance structures.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Consortium agreement and ethics approval confirmed	WP1	M3	Signed consortium agreement; ethics committee approval letter submitted to HaDEA
MS2	Country landscape assessments completed	WP2	M9	Four national landscape assessment reports approved by scientific advisory board
MS3	AI triage module clinically validated	WP2	M15	Validation report with statistical performance metrics

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				submitted to HaDEA
MS4	All eight pilot sites operational and recruiting	WP3	M18	Site initiation reports and first recruitment data confirmed by WP3 lead
MS5	Interim evaluation report and equity analysis	WP4	M30	Interim evaluation report submitted to HaDEA; reviewed by external evaluator
MS6	Scale-up playbook finalised and submitted to EUCanScreen	WP4	M40	Published playbook; acknowledgement of receipt from Joint Action EUCanScreen
MS7	Final conference held and replication commitments secured	WP5	M41	Conference proceedings; minimum 6 signed letters of intent from health authorities

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Project Management and Quality Plan	WP1	M3	Report	Consortium
D1.2	Mid-term and Final Progress Reports	WP1	M42	Report	Public
D2.1	National Landscape Assessment Reports (x4)	WP2	M9	Report	Public
D2.2	AI Decision-Support Tool Validation Report	WP2	M15	Scientific/ Technical Report	Public
D2.3	Integrated Screening Pathway Protocol and Training Materials	WP2	M17	Guidelines/ Protocol	Public
D3.1	Pilot Implementation Report (all 8 sites)	WP3	M37	Report	Public
D4.1	Interim Evaluation and Equity Analysis Report	WP4	M30	Report	Public
D4.2	Final Mixed-Methods Evaluation Report	WP4	M41	Scientific/ Technical Report	Public
D4.3	Scale-Up Playbook and Replication Toolkit	WP4	M40	Toolkit/ Guidelines	Public
D5.1	Communication and Dissemination Strategy	WP5	M3	Plan	Public

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D5.2	Policy Briefs for DG SANTE and National Ministries (x4)	WP5	M38	Policy Brief	Public
D5.3	Final Conference Proceedings and Replication Commitment Report	WP5	M42	Report	Public

3.5 Resources and Budget Summary

Cost Category	Amount (EUR)
Personnel Costs	2850000
Subcontracting	350000
Travel & Subsistence	180000
Equipment	120000
Other Direct Costs	200000
Indirect Costs (25%)	300000
TOTAL	4000000

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	Low community participation in pilot regions due to screening hesitancy or low health literacy, leading to insufficient enrolment to meet primary KPIs.	Medium	High	Early co-design of multilingual and culturally adapted invitation materials with community health mediators and patient organisations; deployment of mobile screening units to reduce access barriers; contingency protocol to extend recruitment window by up to 4 months within the project timeline.
R2	Data governance and GDPR compliance challenges arising from cross-border sharing of pseudonymised screening and AI performance data between four national health systems.	Medium	High	Appointment of consortium Data Protection Officer from month 1; Data Management Plan aligned with GDPR and EHDS interoperability standards; use of federated analysis approaches where cross-border data transfer is not legally permissible.
R3	AI triage module fails to meet pre-specified clinical validation thresholds in one or more national contexts, preventing deployment in pilot	Low	High	Validation conducted in two sites (Italy and Portugal) prior to pilot launch; modular design allows deployment of FIT-based pathway

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	phase.			without AI component if needed, preserving primary screening uptake objectives; fallback to standard triage protocol.
R4	Partner capacity constraints or staff turnover in lower-resource settings (Romania, Lithuania) leading to delays in pilot implementation.	Medium	Medium	Detailed staffing plans with named deputies for all key roles; subcontracting provision for temporary clinical staff support; quarterly management board reviews to identify capacity gaps early.
R5	Regulatory divergence between Member States regarding AI medical device certification (EU AI Act, MDR) delaying AI tool deployment in specific national contexts.	Low	Medium	Legal and regulatory landscape assessed in WP2 landscape mapping (month 1–9); engagement with national competent authorities from project inception; pathway designed to function independently of AI component if regulatory clearance is delayed in any single country.

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